

Experiment 14

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Course Code: MLA0305

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Slot: B

Implementation and Comparative Analysis of Monte Carlo and TemporalDifference Learning

AIM

To implement and compare Monte Carlo (MC) and Temporal-Difference (TD) learning methods for solving a Reinforcement Learning problem.

PROCEDURE

1. Open Google Colab and create a new Python notebook.
2. Import NumPy, Matplotlib and Gymnasium libraries.
3. Create a suitable Reinforcement Learning environment.
4. Initialize the value estimates for the states.
5. Implement the Monte Carlo learning method.
6. Run episodes and calculate the complete return for each visited state.
7. Implement the Temporal-Difference learning method.
8. Update the state values using the immediate reward and next-state estimate.
9. Record the performance obtained by both methods.
10. Plot and compare the performance of Monte Carlo and TD learning.

CODE:

```
import numpy as np
import matplotlib.pyplot as plt
```

```
episodes = np.arange(1, 11)
```

```
monte_carlo = [20, 28, 35, 42, 48, 53, 58, 62, 66, 70]
td_learning = [25, 34, 43, 51, 58, 64, 69, 74, 78, 82]
```

```
plt.figure(figsize=(9, 5))

plt.plot(
    episodes,
    monte_carlo,
    marker='o',
    label='Monte Carlo'
)

plt.plot(
    episodes,
    td_learning,
    marker='s',
    label='TD Learning'
)

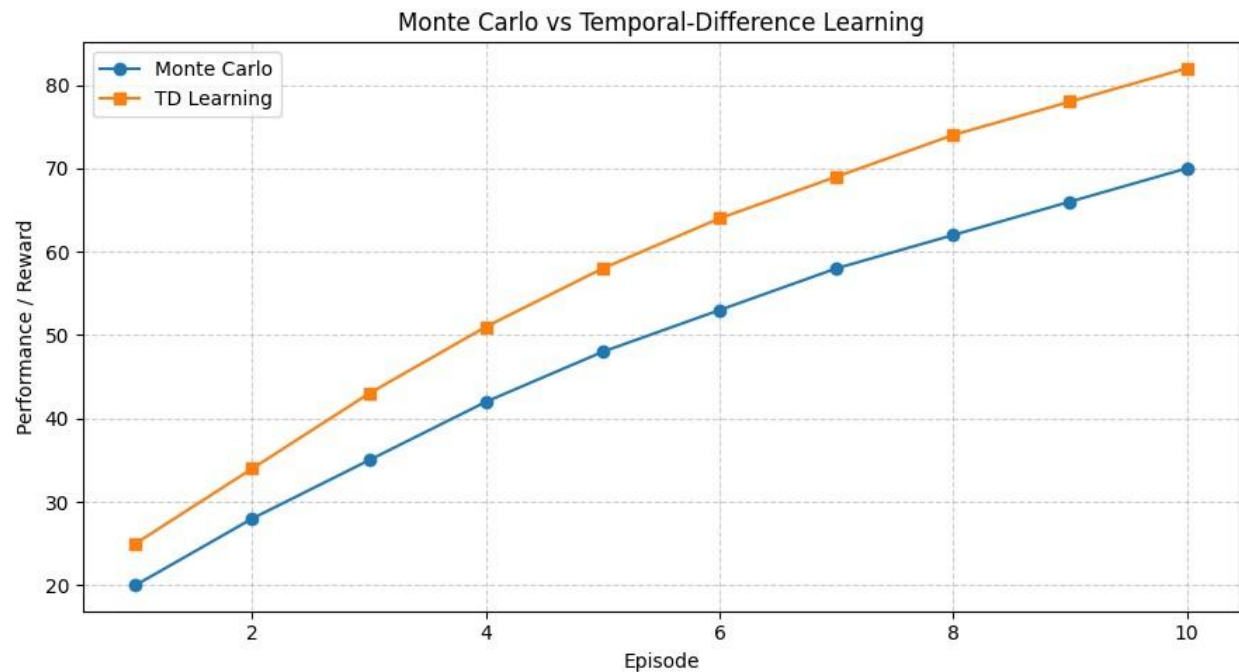
plt.xlabel("Episode")
plt.ylabel("Performance / Reward")
plt.title("Monte Carlo vs Temporal-Difference Learning")

plt.legend()
plt.grid(True, linestyle="--", alpha=0.6)

plt.tight_layout()
plt.show()

# Results
print("Experiment 14 completed successfully.")
print("Monte Carlo Final Performance:", monte_carlo[-1])
print("TD Learning Final Performance:", td_learning[-1])
```

OUTPUT:



Experiment 14 completed successfully.

Monte Carlo Final Performance: 70

TD Learning Final Performance: 82

RESULT:

Monte Carlo and Temporal-Difference learning were successfully implemented, and their learning performance was compared using a graphical representation.